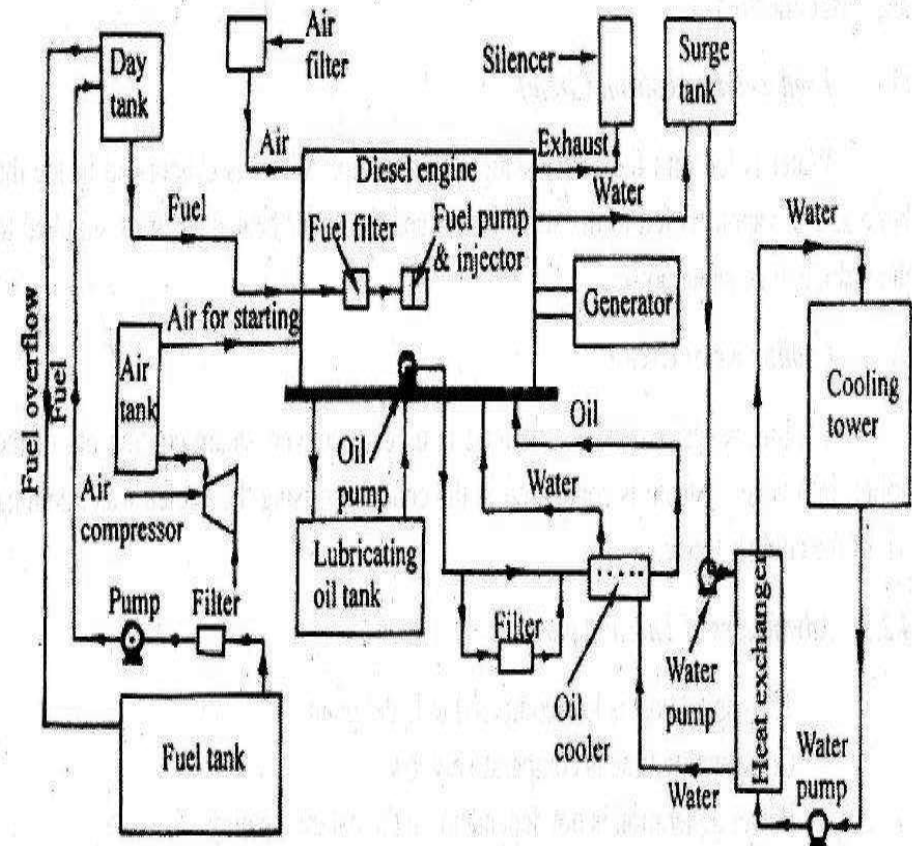


Diesel power plant

- Diesel Engine
- Air intake system
- Fuel supply system
- Exhaust system
- Cooling system
- Lubricating system
- Generator

When the diesel fuel burning inside the engine, its start to produce a mechanical power. The combustion of diesel fuel produces increased temperature & pressure inside the engine. Due to this pressure gases are formed, this gas pushes the piston inside the diesel engine, and then mechanical power is produced. With the use of this mechanical power the shaft of diesel engine starts rotating.



Diesel power plant

(i) Fuel supply system

Fuel from the storage tank is pumped through a filter into a smaller tank called all day tank. This tank supplies the daily requirements of the diesel engine.

(ii) Air intake system

The air required for the combustion of fuel inside the diesel engine cylinder is drawn through the air filter. The purpose of the filter is to remove dust from the incoming air.

(iii) Exhaust system

The exhaust gases coming out of the engine are very noisy. In order to reduce the noise a silencer is used.

(iv) Cooling system

- The temperature of the burning fuel inside the engine cylinder is 1500°C to 2000°C . In order to lower this temperature, water is circulated around the engine.
- The hot water leaving the jacket is passed through the heat exchanger.
- The heat from the heat exchanger is carried away by the raw water circulated through the heat exchanger and is cooled in the cooling tower.

(v) Lubricating system

- This circuit includes lubricating oil tank, oil pump and oil cooler.
- The purpose of the lubrication system is to reduce the wear of the engine moving parts. Part of the cylinder such as piston, shafts, valves must be lubricated.
- Lubrication also helps to cool the engine.

Advantages and disadvantages of diesel power plant

Advantages

1. Very simple design also simple installation and occupies less space.
2. Limited cooling water requirement.
3. Diesel power plant are more efficient than steam power in the range of 150MW capacity.
4. Quickly started and put on load.
5. It can respond to varying loads without any difficulty.
6. Smaller storage is needed for the fuel.
7. Layout of power plant is quite simple.
8. There is no problem of ash handling.
9. Less supervision required

Disadvantages

1. High Maintenance, lubrication cost and operating cost.
2. Fuel cost is more, since in India diesel is costly.
3. The plant cost per kW is comparatively more.
4. The life of diesel power plant is small due to high maintenance.
5. Noise is a serious problem in diesel power plant.
6. Diesel power plant cannot be constructed for large scale.